

SeekPCMdb: knowledge on disease-associated protein-coding mutations

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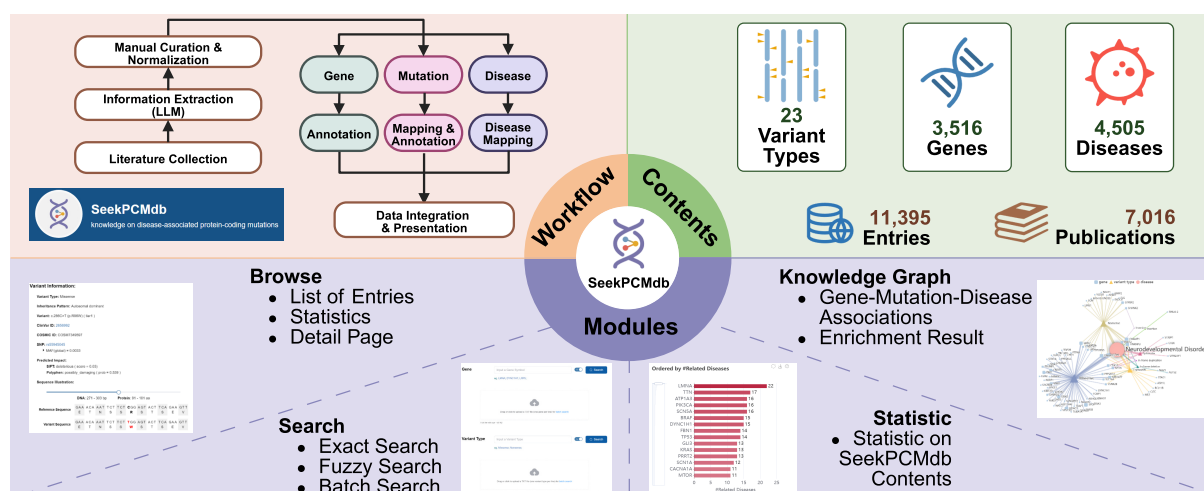
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Abstract

Protein-coding mutations (PCMs) play crucial roles in the onset and progression of human diseases. With the widespread application of next-generation sequencing, an increasing number of disease-associated PCMs have been reported. However, these mutations are scattered across a vast body of literature, limiting their accessibility for systematic investigation. To address this challenge, we present SeekPCMdb (<https://ngdc.cncb.ac.cn/seekpcmdb/>), a literature-derived knowledge base that systematically catalogs human disease-related PCMs. SeekPCMdb employs a semi-automated workflow that integrates manual curation of high-quality publications with large language model (LLM)-assisted information extraction, capturing gene–mutation–disease relationships from >12 000 PubMed abstracts. The current release contains 11 395 curated entries, covering 3516 genes, 4505 diseases, and 23 mutation types. Each entry is richly annotated at multiple levels, including gene function, mutation consequence, and disease classification, enabling thorough biological interpretation. Comparative analysis shows that 63.7% of PCMs in SeekPCMdb are not cataloged in ClinVar, underscoring its unique contribution. SeekPCMdb features a user-friendly web interface that supports interactive searching, visualization, and download, along with multi-dimensional filtering based on gene, mutation type, or disease. By bridging unstructured literature and structured mutation knowledge, SeekPCMdb provides an invaluable resource for investigating genotype–phenotype correlations, assessing mutation pathogenicity, and identifying cross-disease mutational patterns.

Graphical abstract



Introduction

Protein-coding mutations (PCMs) play critical roles in the pathogenesis of various human diseases, especially Mendelian disorders [1–5]. Recent advances in high-throughput sequenc-

ing technologies, particularly whole-exome sequencing (WES) and whole-genome sequencing (WGS), have revolutionized our ability to detect both novel and known pathogenic variants [6–8]. While these studies collectively constitute a rich

Received: August 15, 2025. Revised: October 2, 2025. Accepted: October 17, 2025

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repository of gene–mutation–disease association data that enhances our understanding of disease mechanisms, the predominantly unstructured presentation format of such information poses substantial challenges for systematic integration and computational mining.

Existing databases such as ClinVar [9], HGMD [10], and COSMIC [11] have significantly contributed to the collection of disease-associated variants. These resources provide essential support for variant annotation and clinical decision-making. However, because they rely primarily on manual submissions, expert curation, or contributions from specific research institutions, they often struggle to keep pace with the rapid expansion of the literature. Furthermore, access to some of these resources requires registration, which limits their utility in both research and clinical practice.

To bridge the gap between unstructured literature and structured mutation knowledge, we developed SeekPCMDB, a literature-derived knowledge base focusing on disease-associated PCMs in humans. The database is constructed using a structured, semi-automated workflow that combines manually curated high-quality literature with large language model (LLM)-assisted information extraction techniques. This approach enables the efficient and accurate extraction of gene–mutation–disease triplets from large-scale biomedical texts, followed by standard processing and multi-level annotations.

SeekPCMDB focuses on the roles of PCMs in diseases and supports sequence-level variant mapping, gene function annotation, and disease ontology normalization. Moreover, it incorporates rare disease annotations and Genome-Wide Association Study (GWAS) resources to enrich the clinical and functional context of the mutations. By integrating information on inheritance patterns, disease classifications, and functional impacts, SeekPCMDB aims to serve as a scalable and user-friendly platform for researchers and clinicians, facilitating systematic insights into complex diseases and advancing precision medicine.

Materials and methods

Literature collection

We queried PubMed using the following combination of keywords: (((((Whole exome sequencing) OR (WES)) OR (Whole genome sequencing)) OR (WGS)) AND (mutation)) AND (human), where “mutation” indicated specific mutation types that presumably could result in sequence changes in proteins. To capture a comprehensive set of PCMs, we systematically queried using 10 mutation types: missense, nonsense, *de novo*, frameshift, splice-site, in-frame, stop-loss, start-loss, synonymous, and protein-coding. The number of articles retrieved for each mutation type is summarized in Table 1.

After removing duplicate entries identified through multiple keyword combinations, a total of 12 159 unique publications were retained. We downloaded these records in PubMed format and extracted key metadata fields, including PMID, publication date (DP), title (TI), abstract (AB), and journal title (JT), for downstream mutation information extraction.

AI-assisted data extraction with DeepSeek

To efficiently extract structured gene–mutation–disease associations from the unstructured metadata, we employed DeepSeek [12] for automated information extraction.

Table 1. Literature retrieval statistics by mutation type

Mutation Type	Access Date	# Records
Missense mutation	2025-05-08	5434
Nonsense mutation	2025-05-20	1881
<i>De novo</i> mutation	2025-06-11	3235
Frameshift mutation	2025-07-02	1850
Splice-site mutation	2025-07-02	1035
In-frame mutation	2025-07-02	398
Stop-loss mutation	2025-07-02	24
Start-loss mutation	2025-07-02	11
Synonymous mutation	2025-07-02	879
Protein-coding mutation	2025-07-02	1673
Total		12 159

DeepSeek is an efficient, AI-driven assistant designed to enhance learning, problem-solving, and productivity through conversational interaction. Compared to manual extraction, DeepSeek significantly improves efficiency in identifying key entities from metadata and generating structured outputs. Moreover, DeepSeek provides higher accuracy in establishing correct gene–mutation–disease relationships, particularly when multiple genes and multiple mutations are discussed within a single abstract. We utilized the DeepSeek V3-0324 model deployed on the China Science and Technology Cloud (<https://chat.ai4cas.cn/>) to facilitate continuous extraction without being affected by request throttling or limited server availability.

For each article, the metadata fields (PMID, DP, TI, AB, JT) were submitted to DeepSeek with explicit instructions to extract the following structured information:

- PMID
- Publication date
- Journal
- Reported gene
- Mutation description
- Mutation type
- Inheritance pattern
- Associated disease
- A one-sentence summary of the article’s main finding

If any of the above fields were missing from the metadata, the model was instructed to return a dash (-) as a placeholder.

LLM performance evaluation and confidence classification

To quantitatively assess the performance of DeepSeek in extracting gene–mutation–disease relationships, we randomly sampled 1000 abstracts for evaluation. Extraction outcomes were classified as positive when all three entities were successfully identified and negative otherwise. Each extraction was then cross-validated against the source abstract: records explicitly supported by the text were considered true positives, whereas cases where the abstract did not contain a complete relationship and the LLM likewise did not extract one were regarded as true negatives.

In addition to this evaluation, all extracted records underwent manual review at two stages: an initial plausibility check during extraction and a subsequent verification after aggregation. Discrepancies were resolved by consensus among curators, and synonymous terms were standardized into unified labels.

To improve transparency, we further implemented a two-tier confidence classification. Records classified as Tier 1 represent mutations explicitly reported in abstracts, whereas records of Tier 2 correspond to mutations inferred by the LLM from partial or indirect information (e.g. nucleotide variants deduced from amino acid substitutions). This classification allows users to distinguish directly reported from inferred records and provides a clear indication of reliability.

Gene annotation

We first manually standardized the nomenclature of all gene names to maintain consistency. Next, gene annotation was performed using the human gene information file downloaded from NCBI (accessed 25 June 2025) [13]. For each gene, we annotated the full gene name, known gene synonyms, and chromosomal location. In addition, each gene was linked to its corresponding GeneCards [14] entry to facilitate functional interpretation.

In addition, we assessed the overlap between genes in SeekPCMDB and five reference gene sets: ClinVar genes [9], Online Mendelian Inheritance in Man (OMIM) genes [15], TSGenes (tumor suppressor genes) [16], DNA Damage Repair (DDR) genes [17], and housekeeping genes [18]. To evaluate the statistical significance of the overlap, Fisher's exact test was applied to calculate the enrichment of SeekPCMDB genes in each reference set. The background gene set was derived from the human gene annotation file provided by NCBI, from which 20 595 protein-coding genes were extracted for analysis.

We conducted pathway enrichment analysis on genes whose mutations are associated with the same disease using the R package clusterProfiler [19]. In addition, for each mutation type (with ≥ 10 associated genes), we performed subtype-specific enrichment analyses to investigate potential differential susceptibility of biological pathways to distinct mutation types.

Mutation annotation

To enhance the biological interpretability of each mutation, we mapped all variants to the coding DNA sequences (CDS) and corresponding protein sequences using human reference genomes based on the position information originally reported in the abstracts. The reference sequence data were downloaded via the Table tool from the UCSC Genome Browser [20] (accessed 24 June 2025) using the GRCh38 reference genome and the GENCODE V48 [21] annotation track. Considering the existence of multiple transcript isoforms per gene, we systematically selected the most representative transcript for each mutation through comprehensive comparison of the reported nucleotide/amino acid changes and positional data. We then displayed the corresponding CDS and protein sequence changes with and without the mutation. This dual-level annotation framework allows researchers to visualize mutations in the context of both the DNA sequence and protein sequence levels, thereby enabling straightforward assessment of the functional impact of each mutation, such as premature termination, amino acid substitution, or disruption of protein domains.

To evaluate the potential biological consequences of each PCM, we incorporated functional predictions using SIFT [22] and PolyPhen-2 [23]. These tools provide orthogonal assessments of the likelihood of a given PCM substitution

affecting protein functions, thereby enriching the annotation beyond purely sequence-based mapping.

To further enrich the annotation, we utilized Variant Effect Predictor (VEP) from Ensembl [24] to determine whether each mutation had a corresponding rsID, thereby assessing its annotation information in dbSNP. In addition, we queried the Genome Aggregation Database (gnomAD) [25] to retrieve the minor allele frequency (MAF) of each SNP across populations. Allele frequencies are reported both overall and stratified by major ancestry groups to facilitate assessment of population-specific prevalence and clinical relevance, including Admixed American, African/African American, Amish, Ashkenazi Jewish, East Asian, European (Finnish), European (non-Finnish), Middle Eastern, South Asian, and the remaining.

To assess the novelty and clinical relevance of the mutations cataloged in SeekPCMDB, we performed a systematic comparison with entries in the ClinVar database (accessed 13 July 2025) and the Catalogue of Somatic Mutations in Cancer (COSMIC, accessed 15 September 2025). A mutation was considered present in both databases if it matched the same gene and shared either the same nucleotide and amino acid changes or matched the nucleotide change alone when amino acid data were not available. Overlap analyses were performed to quantify shared and unique content across databases.

Disease annotation

Reported disease names were first manually curated to resolve formatting inconsistencies and unify synonymous terms. These standardized disease names were then mapped to the Disease Ontology (DO) [26] database to retrieve structured annotations, including DOID identifiers, hierarchical classifications, and detailed disease descriptions. DO data were retrieved using the R package ontologyIndex [27] (accessed 25 June 2025). We used the R package fuzzyjoin [28], specifically using the Jaro-Winkler algorithm [29], to compute string similarity to perform fuzzy matching between disease names in SeekPCMDB and those in the DO. All matches were manually reviewed to ensure accuracy and eliminate false positives.

To distinguish rare diseases from common conditions, we downloaded a curated list of rare diseases from the Orphanet [30] database (accessed 7 July 2025) and conducted the same fuzzy matching procedure. All results were subjected to manual verification.

Furthermore, to identify diseases with existing GWAS data, we downloaded the All Studies file from the GWAS Catalog [31] (accessed 10 July 2025). The traits were then matched against diseases in SeekPCMDB using the same fuzzy matching approach.

Database design

SeekPCMDB was deployed on a virtual machine running CentOS 7.9. We used Nginx v1.26.1 (<https://nginx.org/en/>) to provide HTTP and reverse proxy services. MySQL (<https://www.mysql.com>) served as our database engine. The backend RESTful web service was implemented using the Java Spring Boot framework (<https://spring.io/projects/spring-boot>). For the frontend web interfaces, we employed two modern frameworks: Vue3.js (<https://vuejs.org/>) and Vite (<https://vitejs.dev/>). For the user interface component library, we selected Element Plus (<https://element-plus.org>), which offers a comprehensive set of high-quality and customizable components

that enabled us to build a rich and responsive user interface. Additionally, Echarts (<https://echarts.apache.org/>) was used to generate interactive visualization charts.

Database content and usage

LLM performance evaluation

Before large-scale extraction, we evaluated the accuracy of DeepSeek on 1000 randomly sampled abstracts. The model generated 1443 records, comprising 876 positive and 567 negative records. After cross-validation using the source abstracts, it achieved an overall accuracy of 98.9% (1427/1443), with a true positive rate of 99.1% (868/876) and a true negative rate of 98.6% (559/567). These results provide strong evidence of the reliability of LLM-assisted extraction procedure used in SeekPCMDB.

Overview of SeekPCMDB

SeekPCMDB offers a rich resource of disease-associated PCMs. The workflow is shown in Fig. 1A. The current version of SeekPCMDB includes 12 159 articles retrieved from PubMed using ten predefined mutation types as keywords: missense, nonsense, *de novo*, frameshift, splice-site, in-frame, stop-loss, start-loss, synonymous, and protein-coding. From this resource, we successfully extracted structured disease–gene–mutation information from 7016 publications, yielding 11 395 curated records (11 363 Tier 1 records and 32 Tier 2 records). These records represent 11 338 unique disease–gene–mutation triplets and 11 166 distinct gene–mutation pairs, covering 3516 human genes and 4505 diseases. Notably, although we used ten mutation types for our queries, the resulting publications reported a more diverse list of mutation types, totaling 23. Among all mutation types, missense mutations are by far the most common, representing 6508 records (57.2%). This is followed by frameshift (1705, 15.0%), nonsense (1523, 13.4%), and splice-site mutations (916, 8.1%) (Fig. 1B).

For the extracted disease names, we mapped them using a fuzzy search to the Disease Ontology system. A total of 2086 (46.3%) were successfully matched to the Disease Ontology, providing structured disease annotation. Notably, since many of the studies were case reports, the diseases investigated encompassed various subtypes with distinct syndromes. For example, for the Charcot-Marie-Tooth (CMT) disease, we curated 103 records from 64 publications, encompassing 24 subtypes, such as CMT types 2, 2cc, and 4H, among others. Many of the unmapped diseases were due to undiagnosed or undetermined disease names in the original study. For example, Launay *et al.* reported a novel biallelic missense mutation in six individuals “present with a complex hereditary spastic paraplegia (HSP), peripheral neuropathy, verbal apraxia, and intellectual disability” [32]. Although HSP was successfully mapped with a Disease Ontology ID (DOID: 2476), the disease description in this study was very detailed, and we preserved the full description to ensure data integrity. Overall, the curated diseases and the follow-up mapping of disease terms formed a rich and in-depth resource, allowing SeekPCMDB to serve in many aspects.

Notably, during the manual curation and verification process of the extracted records, we identified 69 entries in which the nucleotide and amino acid changes were inconsistent. Upon manual inspection of the corresponding abstracts, we found that 54 entries were correctly extracted and consistent

with the descriptions in the abstract. However, the remaining 15 entries were cases where only the protein-level mutations were reported in the original abstract, and the corresponding nucleotide changes were inferred by DeepSeek during information extraction. These inferred variants were incorrect, implying a potential limitation of automatic nucleotide-level mutation inference when explicit sequence information is lacking in the source text.

Functional characteristics of items in SeekPCMDB

Mutations

Functional impact predictions revealed that 87.0% of PCMs (5192 out of 5971 successfully mapped mutations to SIFT) were classified as “deleterious” by SIFT, while 76.8% (4136 out of 5388) were predicted as “probably damaging” or “possibly damaging” by PolyPhen-2. A total of 3968 mutations were predicted to be deleterious by both tools, representing a high-confidence subset likely to disrupt protein functions. These results highlighted that although not all PCMs alter protein function, a substantial proportion cataloged in SeekPCMDB may have deleterious effects and therefore warrant further investigation.

Cross-referencing with dbSNP revealed that 57.6% of mutations in SeekPCMDB have matched records in dbSNP, i.e. with an rsID. Among these SNPs, 97.0% had MAF < 0.05, indicating that although they were reported as SNPs, they are rare in the general population. Cross-referencing with ClinVar revealed that 36.3% (4136 out of 11 395) of mutations in SeekPCMDB were also present in ClinVar, while the remaining 63.7% were unique to SeekPCMDB. A comparison with COSMIC further identified 323 overlapping entries between the two databases, whereas the vast majority of entries in each resource were unique. This underscored that SeekPCMDB captured a substantial number of mutations not currently included in public clinical variant databases. Thus, by systematically extracting mutation data from literature abstracts, SeekPCMDB presents as a valuable complementary resource to established databases such as ClinVar.

These mutations were mainly categorized into five inheritance patterns, including autosomal dominant (AD, 23.1%), autosomal recessive (AR, 49.2%), X-linked (5.5%), mitochondrial (0.2%), and somatic mutations (1.6%). In addition, some records were *de novo* mutations, including *de novo* AD and *de novo* AR.

Genes

Given that SeekPCMDB focuses on protein-coding mutations, which tend to have significant biological effects, we next compare the genes in SeekPCMDB with those from functionally important gene sets, such as genes from OMIM, housekeeping genes, and tumor suppressor genes. OMIM is an online catalog of human genes and genetic disorders. So far, OMIM has collected over 16 000 genes associated with genetic disorders. Interestingly, 3309 (94.1%) of SeekPCMDB genes have been reported by OMIM, representing a significant over-representation. Although SeekPCMDB does not specifically focus on cancer, we compared our genes with tumor suppressor genes (TSGenes) because the latter tend to be interrupted by loss-of-function mutations, according to the two-hit theory. Only 8.8% of SeekPCMDB genes were previously reported as TSGenes, though this overlap was already statistically over-represented ($P = 7.7 \times 10^{-33}$). Similarly, DDR

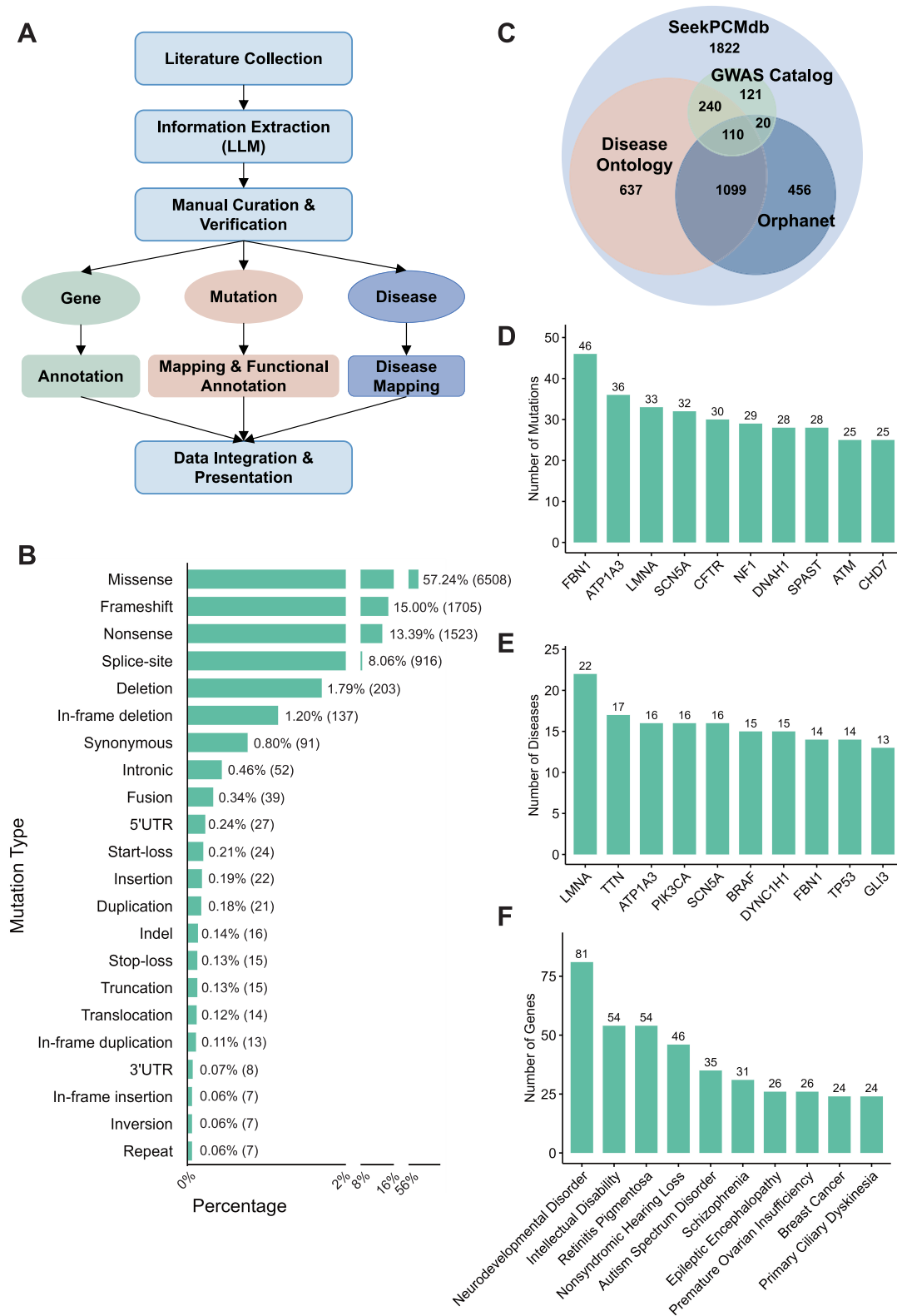


Figure 1. Overview of SeekPCMDb. **(A)** The workflow of data processing. **(B)** Proportional distribution of different mutation types in SeekPCMDb. **(C)** Annotation of diseases in SeekPCMDb using Disease Ontology, Orphanet, and the GWAS Catalog. The Venn diagram illustrates the overlap and complementarity among the three resources in disease annotation. **(D)** The 10 most frequently reported genes ranked by their unique mutations. **(E)** The 10 genes most frequently linked to diverse diseases. **(F)** The top 10 diseases associated with the largest number of genes.

Table 2. Characteristics of the records in SeekPCMDB

Category	# Total	# Overlap	Fisher p-value	Percentage
Genes	3516			
Overlap with ClinVar	19 250	3330	2.7×10^{-138}	94.7%
Overlap with OMIM genes	17 451	3309	9.0×10^{-296}	94.1%
Overlap with TSGenes	1217	309	7.7×10^{-33}	8.8%
Overlap with DDR genes	219	78	9.2×10^{-13}	2.2%
Overlap with housekeeping genes	3804	636	0.008	18.1%
Diseases	4505			
Overlap with DO	14 387	2086	–	46.3%
Overlap with Orphanet	11 239	1685	–	37.4%
Overlap with GWAS Catalog	22 198	491	–	10.9%

#TSGenes: tumor suppressor genes; DO: Disease Ontology; DDR: DNA Damage Repair.

genes are also likely interrupted by loss-of-function mutations and are statistically significantly associated with our SeekPCMDB genes ($P = 9.2 \times 10^{-13}$). Finally, we compared our genes with housekeeping genes, which are known for their critical functions in biological processes. As shown in Table 2, 18.1% of SeekPCMDB genes are also housekeeping genes ($P = .008$), implying their functional importance. Taken together, these results indicated that SeekPCMDB preferentially captured genes with disease relevance or essential cellular functions.

Diseases

We next compared the diseases in SeekPCMDB with those from the Orphanet database, which represents a collection of rare diseases, and those from the GWAS Catalog, which mainly includes common diseases that have been investigated for their associated common variants. As expected, the diseases in SeekPCMDB shared more with the Orphanet database (1685 diseases, 37.4%) than with the GWAS Catalog (491 diseases, 10.9%) (Fig. 1C and Table 2).

Disease–gene–mutation knowledge graph

To enable intuitive visualization and exploration of associations, we constructed a mutation–gene–disease knowledge graph for all the SeekPCMDB records. Each entry was represented as a triplet, including genes, mutation types, and diseases, with links denoting reported associations between the mutation and the disease. The resulting knowledge graphs were visualized using two approaches: (i) Sankey diagrams for summarizing the flow from genes to mutation types and associated diseases and (ii) interactive network graphs for dynamic exploration of connections among entities, with the complete network graph shown in Fig. 2A.

The gene with the largest number of mutations is *FBN1*, with 46 entries spanning 7 mutation types and associated with 14 distinct diseases (Fig. 1D). In comparison, *LMNA*, which encodes lamin A/C, is linked to the largest number of diseases (22 in total) (Fig. 1E). These disorders span four major categories: muscular dystrophies, cardiac disorders, lipodystrophic syndromes, and premature aging disorders. Mutations in *LMNA* comprise four distinct types: missense, nonsense, frameshift, and splice-site. In terms of disease, neurodevelopmental disorders are associated with the largest number of genes in SeekPCMDB, involving 81 genes and 8 mutation types (Fig. 1F). These genes are significantly enriched in pathways related to synaptic function, neurotransmission, and neuronal signaling. Genes within different biological pathways may be differentially susceptible to specific types of mutations. For in-

stance, genes with missense mutations are predominantly associated with neuronal signaling and hormone-related pathways, such as regulation of presynaptic membrane potential and intracellular steroid hormone receptor signaling. In contrast, genes with nonsense mutations are more enriched in cell division and cytoskeletal organization, including mitotic cytokinesis and protein localization to microtubules.

Furthermore, we observed that 2074 genes were each associated with only a single disease, while 3796 diseases were linked to only one gene. These findings highlight the presence of numerous isolated nodes within the network, suggesting that a substantial portion of gene–disease associations in SeekPCMDB are disease-specific. This duality of highly connected hubs and numerous isolated nodes underscores the complexity of the genetic landscape captured in SeekPCMDB.

Case applications using SeekPCMDB

The records in SeekPCMDB span a wide range of diseases, from Mendelian disorders to common complex diseases. We chose two representative disorders to demonstrate the value of SeekPCMDB. The first application is hearing loss (HL). HL is a heterogeneous condition with partial or complete hearing impairment. It can be generally categorized as nonsyndromic HL (NSHL) and syndromic HL (SHL), or hereditary HL (HHL) and *de novo* HL. Using “hearing loss” and “deaf” as the keywords, we found 300 records from 180 publications. Among the diseases, NSHL was the most prevalent, with subtypes such as autosomal dominant NSHL (DFNA), autosomal recessive NSHL [33], and X-linked NSHL [34]. In addition, we found many variations of the disease with different syndromes, such as congenital deafness with vestibular areflexia [35], leukodystrophy with ataxia and deafness [36], and progressive cone-rod dystrophy and early-onset sensorineural HL [37]. Previous studies have reported >60 genes associated with HL, whereas SeekPCMDB curated 104 genes. The genes with the largest number of mutations included *KCNQ4*, *MYO15A*, *CDH23*, *GJB2*, and *PKHD1L1*, all of which were widely investigated in HL. The rich information about diseases and genes presented valuable resources for future studies.

The second example is autism spectrum disorder (ASD). ASD is a highly heritable yet heterogeneous neurodevelopmental disorder. Multiple GWASs have been conducted to identify common genetic risk variants associated with ASD. SeekPCMDB collected PCMs occurring in ASD patients, reporting 99 records, 65 genes, and 67 publications. To compare the two lines of evidence at the gene level, we calculated gene-based *P*-values based on the ASD GWAS summary statistics

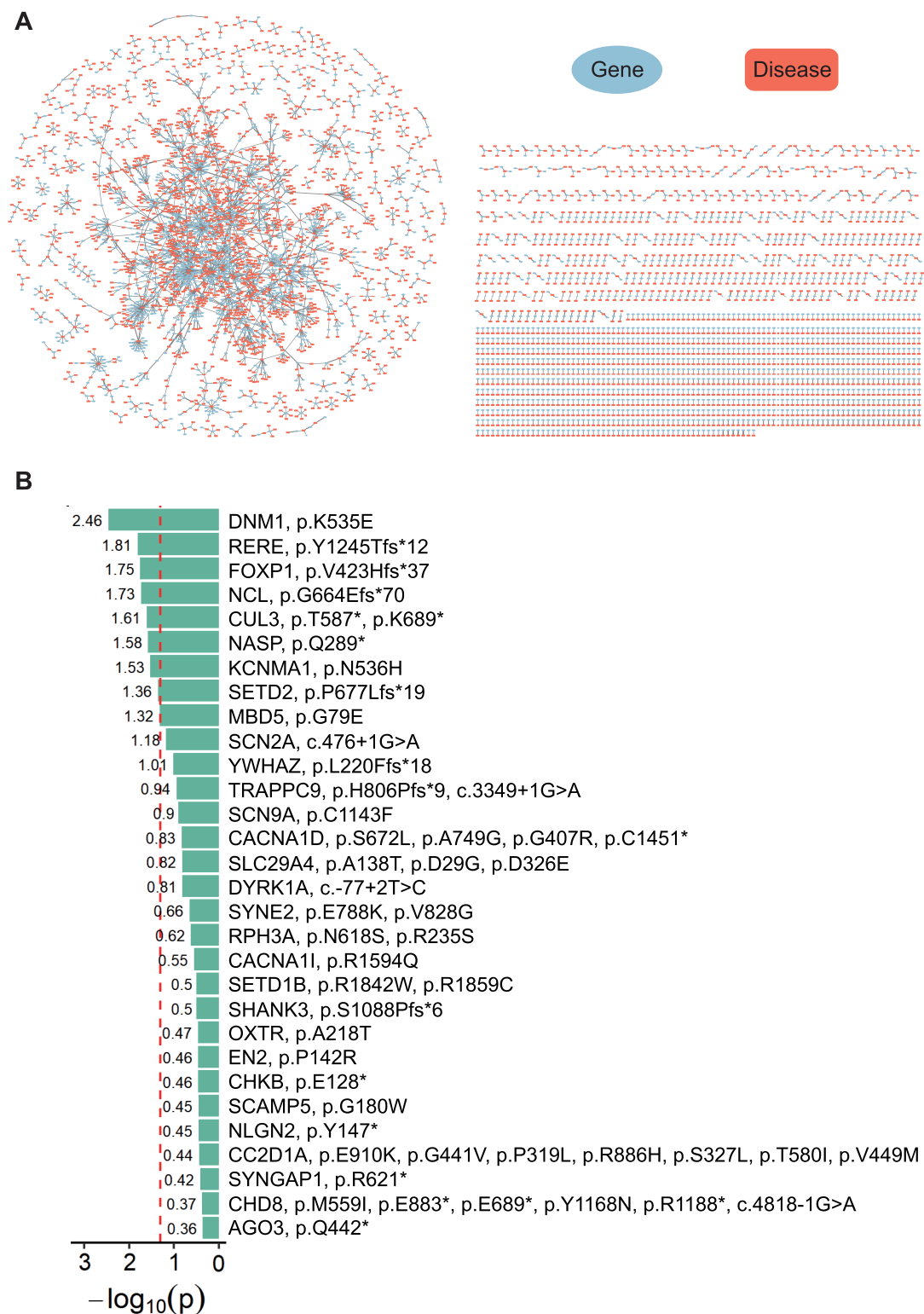


Figure 2. Usage of SeekPCMDB. **(A)** The complete knowledge graph constructed using all records in SeekPCMDB, where blue ellipses denote gene nodes, red rectangles denote disease nodes, and the edge thickness is proportional to the number of gene–disease associations. **(B)** Example analysis for autism spectrum disorder (ASD), illustrating protein-coding mutations (PCMs) and SNP effects of ASD-associated genes. The right panel shows the distribution of PCMs for each gene. The left panel presents the $-\log_{10}(P\text{-values})$ for gene–ASD association derived from MAGMA analysis. The dashed line indicates the significance threshold at $P = 0.05$. This panel shows the top 30 genes ranked by the gene-based $P\text{-values}$ calculated by MAGMA.

[38] using the method MAGMA [39]. As shown in Fig. 2B, none of the PCM genes reached the genome-wide significance level, though nine genes had nominally significant *P*-values: *DNM1* (MAGMA *P* = .0035), *RERE* (*P* = .015), *FOXP1* (*P* = .018), *NCL* (*P* = .019), *CUL3* (*P* = .024), *NASP* (*P* = .027), *KCNMA1* (*P* = .03), *SETD2* (*P* = .044), and *MBD5* (*P* = .048). Thus, the genes collected by SeekPCMDB provided an alternative and complementary line of evidence for the gene and phenotype associations.

Web interface

SeekPCMDB provides an intuitive, user-friendly web interface for facilitating efficient browsing and searching of specific entries. The web interface is organized into seven functional modules: Home, Browse, Search, Statistics, Knowledge Graph, Case Studies, and Documentation (Fig. 3).

The Home page supports three primary query modes, allowing searches by using gene symbol (e.g. *LMNA*), variant type (e.g. missense), or disease name (e.g. breast cancer) (Fig. 3A). We have pre-loaded all available terms for each query option and implemented an intuitive auto-complete dropdown, which offers suggestions for selection and enables faster access to relevant data. Additionally, the homepage features word cloud visualizations that allow users to navigate directly to detailed pages by clicking on a gene or disease term.

The Browse page presents all entries in the database in a structured tabular format, along with key metadata including gene symbol, variant, variant tier, variant type, inheritance pattern, associated disease, and PubMed ID (Fig. 3B). Users can apply advanced search functions, supporting both exact and fuzzy matching, and also batch searches where multiple keywords can be entered separated by the “|” symbol, to efficiently filter results. All displayed data are freely downloadable without the requirement of registration. Additionally, the Statistics section in the Browse page summarizes the current results using interactive charts, including bar charts for the most frequently implicated genes and diseases and pie charts illustrating the distribution of variant types and inheritance patterns (Fig. 3C).

Users can navigate to a more focused page with detailed and full information by clicking any keyword (e.g. Gene, Disease, or PubMed ID) in the Datasets section. The full-information page illustrates the relationships among genes, variant types, and diseases using interactive network graphs and Sankey diagrams (Fig. 3D). In addition, it provides a comprehensive list of annotations on the related genes, variants, diseases, and associated publications, along with intuitive visualizations of mutation consequences, including their genomic positions and protein sequences, and the corresponding reference and altered sequences (Fig. 3E). Additional filtering options are available in the sidebar to refine results based on gene symbol, variant, variant type, disease term, or PubMed ID.

The Search page provides multiple search functions for users to identify genes, variant types, diseases, or PubMed IDs, supporting both exact and fuzzy matching. In addition, a batch search function has been implemented, allowing users to either upload a .txt file with one keyword per line or input multiple keywords separated by the “|” symbol. Batch search supports exact matching only.

The Statistics page contains four sections summarizing genes, variant types, inheritance patterns, and diseases (Fig. 3F). In each section, keywords are ranked by the num-

ber of associated terms and visualized through interactive bar and pie charts. An additional scatter plot displays the distribution and relationships among these categories, with jittering applied to mitigate node overlap and facilitate exploration.

The Knowledge Graph page provides an interactive network visualization platform. Users can query genes or diseases of interest via a search bar, and the system will dynamically render all associated genes, mutation types, and diseases in both a network graph and a Sankey diagram, revealing their complex interrelationships (Fig. 3G). In addition, for diseases associated with ≥ 10 genes, the page also provides the results of GO enrichment analysis. Users can navigate to the GO term's page by clicking on a node in the graph or the corresponding GOID in the statistics table below (Fig. 3H).

The page “Case Studies” was designed to showcase practical applications of SeekPCMDB, using two representative examples, i.e. HL and ASD (Fig. 3I). Each case study integrated three components: (i) a summary panel highlighting the number of genes, variant types, diseases, and publications; (ii) interactive bar charts ranking diseases and genes by their respective record counts; and (iii) a dedicated table presenting all case-specific entries with associated metadata.

Finally, the Documentation page outlines the data acquisition and curation workflow, summarizes database content, and includes a detailed user guide to assist researchers in navigating and utilizing this resource.

Database update

To ensure the timeliness and completeness of SeekPCMDB, we will update the database regularly. Specifically, a comprehensive update will be conducted every three months, including re-querying PubMed to incorporate newly published studies, particularly those reporting novel mutations or focusing on Mendelian diseases. In parallel, we will continuously monitor the development of leading large language models, such as DeepSeek and GPT, and accordingly update our information extraction pipeline based on performance evaluations. This approach aims to improve both the accuracy and the coverage of the extracted data. Additionally, we will periodically update external reference databases, including NCBI gene annotation files, ClinVar, and the Disease Ontology, to ensure that the annotations remain consistent with the current version. To enhance the interactivity and openness of the database, SeekPCMDB encourages users to share suggestions, report potential issues, or recommend new data sources through the “Contact” feature on the website, thereby contributing to the continuous improvement of the resource.

Discussion and future developments

SeekPCMDB is an open-access, integrated resource dedicated to the study of disease-associated PCMs in humans. It employs a semi-automated pipeline that combines manual curation with information extraction powered by large language models, systematically retrieving gene–mutation–disease associations scattered throughout biomedical literature abstracts and converting them into standardized formats. To the best of our knowledge, SeekPCMDB is the first database to apply LLMs for extracting mutation-related information, significantly enhancing the efficiency of PCM curation compared to fully manual approaches.

One of the major strengths of SeekPCMDB lies in its multi-level annotation system, which provides rich contextual

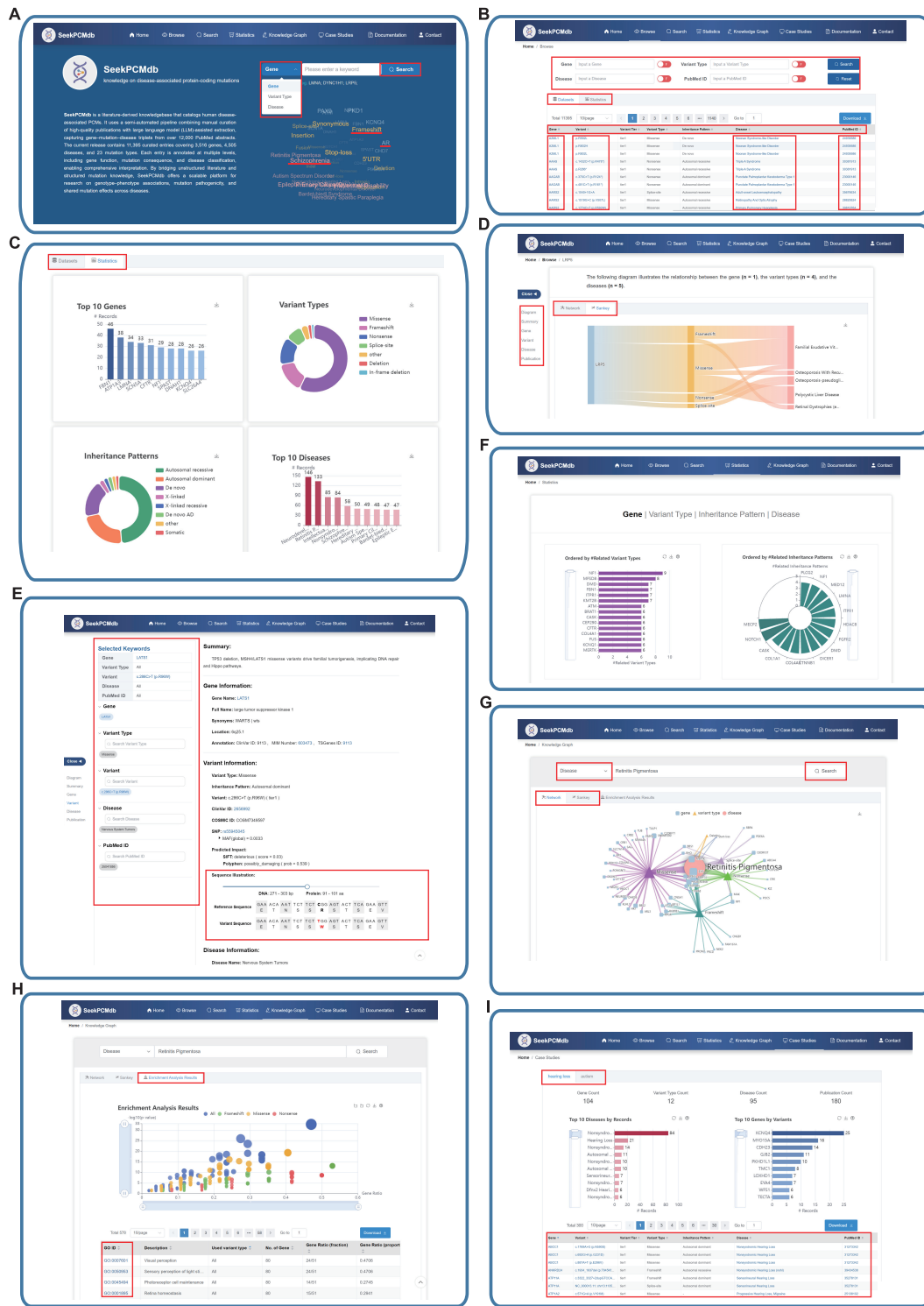


Figure 3. Screenshots of web pages for SeekPCMDB. **(A)** The home page supports three search modules that enable users to efficiently access data in SeekPCMDB. **(B)** The browse page enables users to retrieve data using the built-in search tools and to access genes, variant types, diseases, and other detailed information in a tabular format. **(C)** The statistics section of the browse page provides interactive charts that summarize the distribution of genes, variant types, diseases, and inheritance patterns. **(D)** An interactive Sankey diagram on the full-information page illustrates gene–variant type–disease associations. **(E)** The main content section of the full-information page displays comprehensive information, including a visualized sequence viewer. A collapsible filter panel is provided on the left side. **(F)** The statistics page presents interactive bar, pie, and scatter charts summarizing the distributions and relationships of genes, variant types, inheritance patterns, and diseases, allowing users to explore the data by selecting different keywords. **(G)** The knowledge graph page offers interactive network and Sankey diagram visualizations, dynamically rendering associations among genes, variant types, and diseases. **(H)** The results of GO enrichment analysis for disease-associated genes are presented as interactive scatter plots and tables on the knowledge graph page. **(I)** The case studies page presents representative examples of HL and ASD, with statistical summaries, interactive visualizations, and tabular displays of related entries.

information for each “gene–mutation–disease” triplet. At the gene level, annotations enable users to explore gene functions and assess pathway enrichment across disease-associated gene sets. At the mutation level, variants are mapped to the corresponding genomic and protein sequences and further annotated with functional impact predictions using SIFT and PolyPhen-2. This allows users to intuitively visualize nucleotide and amino acid changes and simultaneously assess their potential effects on protein functions. At the disease level, comprehensive annotations are provided using Disease Ontology, supplemented by Orphanet for rare disease classification and GWAS Catalog for GWAS resources. While only 46.3% of diseases could be mapped to DO, additional mappings to Orphanet and GWAS Catalog extend structured classification and enhance interpretability across the database. This multi-dimensional annotation framework greatly enhances the interpretability and utility of PCM data in both clinical and systems biology contexts.

Because many annotations in SeekPCMDB are derived from external databases (e.g. ClinVar, COSMIC, Disease Ontology, and Orphanet), their accuracy is partially dependent on these sources. To address this, the version and access date of each resource are provided, and the database will be regularly updated to synchronize with these external references, enabling users to assess potential discrepancies due to updates or inconsistencies.

When benchmarked against existing variant resources, SeekPCMDB demonstrates both complementary strengths and coverage gaps. ClinVar compiles clinically interpreted variants from laboratories and expert panels. SeekPCMDB aggregates literature-derived mutations, including recent large-scale sequencing reports. The two databases share only 36.3% of entries, with 3 592 675 ClinVar variants absent from SeekPCMDB. A direct large-scale comparison with HGMD was not feasible due to subscription restrictions. To partially address this, we compared SeekPCMDB with COSMIC, an open-access repository of cancer-associated somatic mutations, revealing 323 overlapping entries and 1 179 850 COSMIC variants absent from SeekPCMDB. These differences largely reflect the differences in the scope and curation strategies: ClinVar aggregates clinically interpreted variants submitted by laboratories and expert panels, COSMIC focuses on cancer-specific somatic variants, and SeekPCMDB employs LLM-assisted extraction of disease-associated mutations from PubMed abstracts. Collectively, these comparisons highlight SeekPCMDB's complementary role in capturing literature-reported variants not yet incorporated into existing clinical or cancer-focused databases.

SeekPCMDB is also to be distinguished from OMIM, Orphanet, and Disease Ontology. OMIM emphasizes gene–disease relationships. Orphanet and Disease Ontology focus on rare diseases or structured disease terms. In contrast, SeekPCMDB provides detailed variant-level annotations and functional impact predictions for each disease–gene–mutation triplet. Additionally, SeekPCMDB integrates a knowledge graph linking genes, mutations, and diseases, enabling cross-disease mutation pattern analyses.

Nonetheless, SeekPCMDB has several limitations. Currently, it extracts information solely from PubMed abstracts. While most studies report key PCMs in the abstract, mutations mentioned only in the full text or supplementary materials may be missed. Incomplete information in abstracts can affect LLM extraction accuracy; to mitigate this, a tiered con-

fidence system was applied to all records, providing users with an indication of reliability. As LLMs continue to improve in both processing capabilities and input length, future iterations of SeekPCMDB are expected to support full-text extraction beyond abstracts. Moreover, human review inevitably introduces potential curator bias and may constrain scalability. Future versions will explore systematic approaches such as cross-validation with multiple LLM prompts and automated rule-based filters to further improve accuracy and scalability. Another limitation is that the literature retrieval strategy primarily targets large-scale sequencing studies (e.g. whole-exome or whole-genome sequencing). Although this approach captures the majority of PCMs, clinically relevant variants reported in case studies or targeted sequencing projects may still be underrepresented. Future updates of SeekPCMDB will incorporate broader keyword strategies to improve coverage. A further limitation is that current functional annotation is primarily based on sequence-level predictions. While SIFT and PolyPhen-2 provide valuable evidence of potential functional impact, they do not capture structural alterations. Incorporating three-dimensional analyses, for example by leveraging AlphaFold, represents a promising direction for future updates. Such structure-based annotations would allow researchers to explore conformational changes, domain disruptions, or altered binding interfaces caused by PCMs.

Overall, SeekPCMDB represents a valuable resource for researchers investigating genotype–phenotype relationships, mutation pathogenicity, and shared mutation effects across diseases.

Acknowledgements

We would like to thank Ms. Shuhua Li for helping improve the graphic abstract. We also sincerely acknowledge all developers, curators, and maintainers of the public databases, tools, and resources used in this study for providing invaluable resources and support that enabled the construction and annotation of SeekPCMDB.

Author contributions: Peng Yang (Data curation [lead], Formal analysis [lead], Methodology [lead], Visualization [equal], Writing—original draft [lead]), Ziyang Wang (Software [lead], Visualization [equal], Writing—original draft [supporting]), Tuo Xiong (Software [lead]), Zhanjie Fang (Data curation [supporting]), Xiuli Zhu (Data curation [supporting]), Lilin Yao (Data curation [supporting]), Ying Cui (Data curation [supporting]), Rongxing Wei (Data curation [supporting]), Zhiming Qian (Data curation [supporting]), Jian Wang (Resources [supporting]), Yin-Ying Wang (Supervision [supporting]), and Peilin Jia (Conceptualization [lead], Funding acquisition [lead], Methodology [lead], Project administration [lead], Resources [lead], Supervision [lead], Writing—review & editing [lead]).

Conflict of interest

None declared.

Funding

This research was supported by the Strategic Priority Research Program of the Chinese Academy of Sciences (XDA0460202 and XDA0460204), the National Natural Science Foundation of China (92374103, 32270706, 62473355),

and the National Science and Technology Major Project (2024ZD0533502). Funding to pay the Open Access publication charges for this article was provided by the Strategic Priority Research Program of the Chinese Academy of Sciences (XDA0460202).

Data availability

SeekPCMDb is freely available at <https://ngdc.cncb.ac.cn/seekpcmdb/>. Users can access resources without registration or login.

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